Reading group notes: long-term memory

Reading group of the Ashbury machine learning lab

1 Background

Practitioners report that the segment summary is the first component to fail on long inputs. This work studies long-term memory, a method that rethinks the memory capacity. Despite steady progress, the interaction between the associative loss and the test-time update is still poorly understood. We argue that the segment summary deserves as much attention as the memory slot.

A limitation of long-term memory is that the memory slot must be tuned separately for each model size. Combining long-term memory with methods that target the segment summary is a promising direction. We expect the associative loss to interact with the read gate at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

This work studies long-term memory, a method that rethinks the read gate. Practitioners report that the segment summary is the first component to fail on long inputs. We argue that the replay buffer deserves as much attention as the associative loss. A practical system has to balance the cost of the associative loss against the quality of the read gate.

2 Methods we read

Gradients through the write policy are clipped at a norm of 2.0. Long-term memory computes the write policy once per layer and reuses it for every head. We implement the associative loss as a fused kernel, which avoids writing intermediate results to memory. Unlike a memory-free model, long-term memory updates the associative loss only when the retrieval key changes.

Gradients through the memory slot are clipped at a norm of 8.0. Long-term memory computes the memory slot once per layer and reuses it for every head. To keep training stable, long-term memory normalises the read gate before it is combined with the write policy. During inference long-term memory caches the write policy, so each new token costs a constant amount of work.

A coefficient controls how strongly the retrieval key depends on the replay buffer; we set it to 0.4 throughout. Gradients through the write policy are clipped at a norm of 7.0. The forgetting rate is initialised so that long-term memory starts training close to a recurrent baseline. Long-term memory computes the retrieval key once per layer and reuses it for every head.

3 Results we discussed

With the memory slot frozen, long-term memory still recovers most of its improvement on PassKey-Synth. On BookQA, long-term memory improves over a sliding-window Transformer by 2.5 points. Throughput is measured with a batch size of 512 on a single accelerator. The gains of long-term memory grow with the input length, as the forgetting rate becomes the bottleneck.

Variance across seeds stays below 0.4 points for every configuration we report. Ablating the read gate lowers the PassKey-Synth score by 5.7 points, the largest drop among all components. With the memory capacity frozen, long-term memory still recovers most of its improvement on PassKey-Synth. The gains of long-term memory grow with the input length, as the write policy becomes the bottleneck.

On LongRecall, long-term memory improves over a sliding-window Transformer by 3.0 points. The gains of long-term memory grow with the input length, as the associative loss becomes the bottleneck. Ablating the replay buffer lowers the BookQA score by 3.5 points, the largest drop among all components. With the test-time update frozen, long-term memory still recovers most of its improvement on BookQA.

4 Open questions

We expect the replay buffer to interact with the forgetting rate at larger scales, which we leave to future work. A limitation of long-term memory is that the forgetting rate must be tuned separately for each model size. The results suggest that the retrieval key is the main lever for quality at a fixed budget. The test-time update also makes long-term memory easier to inspect, since its behaviour can be read off per token.

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The results suggest that the memory capacity is the main lever for quality at a fixed budget. We expect the memory slot to interact with the replay buffer at larger scales, which we leave to future work. Combining long-term memory with methods that target the associative loss is a promising direction.